

SAD Lab 4 Section B – Facility Reservation & Approval System

Required Submission Report (Items 22–32) | Course: SAD Lab 4 | Project: Role-Based Campus Facility Reservation and Approval System

Roles: Administrator, Facility Staff, Requester (Student / Faculty / Student Org), Security, Management

22. Stakeholder Statement Analysis

Each stakeholder's concern is translated into an implied requirement that the system must satisfy.

#	Stakeholder	Statement / Concern	Implied Requirement
S1	Student / Faculty / Student Org (Requester)	"I want to check facility availability first, then book it for my activity and know the outcome of my request."	Search availability (FR-01); submit reservation (FR-02); be notified of outcome (FR-09).
S2	Facility Administration	"I need to review every request, approve or reject it, and keep facility information accurate."	Approval workflow (FR-05); per-facility approval policy (FR-04); facility management (FR-10).
S3	Facility Staff	"I need to confirm when a facility is actually used, record completion, and file maintenance concerns."	Usage confirmation / no-show / completion (FR-06); service requests.
S4	Campus Security	"I must know what is authorized after hours and approve large events for safety."	After-hours authorization list (FR-07); dual approval for large events (CR-01).
S5	Management	"I need utilization data to plan capacity and operations."	Utilization reports with CSV export (FR-08); audit-log visibility.
S6	IT / Auditor	"Every decision and data change must be traceable."	Full audit trail (BR-B4-10); read audit logs.

23. Requirements Problems and Clarification Questions

#	Problem / Ambiguity	Clarification Question
P1	"After-hours" is configurable but not confirmed.	Is after-hours = weekends + 18:00–07:00, or only nights / holidays? Who authorizes exceptions?
P2	No maximum booking duration or advance lead-time exists.	What is the max hours per booking and how far in advance can a reservation be made?
P3	Notifications are in-app only.	Is an email / SMS gateway required, or is in-app sufficient for the lab?
P4	No-show handling is defined but has no penalty.	Should repeated no-shows restrict a requester's future bookings?
P5	Auto-confirm facilities skip admin review.	Should auto-confirmed requests still appear in an audit log and be cancelable by admin?
P6	Large-event threshold is set at 100 pax.	Is 100 the agreed threshold, or should it be per-facility?
P7	Recurring events are unsupported.	Do requesters need recurring / series bookings in this release?

24. Stakeholder/Actor Analysis

Actor	Type	Description	Goals	Key Interactions
Requester	Primary	Student, faculty member, or student organization representative	Search availability, submit requests, track / cancel own bookings	UC-01, UC-02, UC-04, UC-09
Administrator	Primary	Facility Administration / IT Personnel	Manage facilities & users, approve / reject, run reports, audit	UC-03, UC-04, UC-05, UC-07, UC-08, UC-10
Facility Staff	Supporting	Operations staff	Confirm usage, record completion / no-show, service requests	UC-04, UC-06
Security	Supporting	Campus security officer	Approve large events, view after-hours authorization	UC-06, UC-10
Management	Secondary	Facilities Manager	View utilization reports and audit logs	UC-05

25. Ten Functional Requirements (FR)

ID	Requirement
FR-01	The system shall allow any logged-in user to search facility availability by date / time, minimum capacity, and facility type, returning only Active facilities (BR-B4-01).
FR-02	The system shall let a Requester submit a reservation request with facility, purpose, participant count, start, and end; it must reject invalid time ranges (BR-B4-02), capacity overflow, and overlapping schedules (BR-B4-03).
FR-03	The system shall enforce the reservation status workflow: Submitted → Pending → Approved/Rejected, then Scheduled → In Use → Completed, plus Cancelled and No-show (BR-B4-05..07).
FR-04	The system shall apply a per-facility approval policy: Requires approval or Auto-confirm (auto-confirm sets status directly to Approved).
FR-05	The system shall allow only the Administrator to approve or reject non-large requests and shall record the decision (BR-B4-04, BR-B4-10).

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„ŸŃW□œ~□A9ĀôŸ„¹n"¹z%◊)1~ăİŌŸİj~□İ
ôv£İăİŌ5ª»\$¹=□□bŮôHYŌ~KĀăşP,◊□□_1K□□Ÿİ¹µ□P8◊ôM3,ZŸ×dŸªªç"zŌŮ□9◊□ª5mm4Ůð±2ă2×8·K□WēşĒ-n%MŌ□~nİăİ{◊□þ\$ŸwTEÇĒē»9C
İ«◊+þBă◊İ□s◊CŮSŌĐ£İŮðİªç◊□»...±µĒĀŌŃŃŃİ¹Ů½j)ŮŮăĒ·FşēŮ;İŸă~ē±~.İ
"Ēó~Jş·PĀă)qµñøēİ□Ńj¼□W□ĒÆ1>„Đ3□İþŸİĐĒē~"ôhŮ"şş◊w□FG'çSđæ¼¼™3Q□□"òTĐc×±...ŃİİŸŸ-ŌŞ◊İBEİ□+~
{Ů□□İrµñİ½◊İ"Kşæ"◊İ□□ŌH"=F□□ŌĐē:ē²~"◊%◊◊ĐİVE^Ůİ'Ůt^◊Đž;ĐžŮŮ□/Æ□□K□mĐTđŌ£İf~
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ª/wěăXg,V&~ªmĀĐCE^mrncē—æTTŸ□□ēİ◊İŮvc¾¼◊◊µçâ□□,□□□9İ(İªrİ)7Uđ·nÇ/İÇ=□□P"ôþç◊ªqđ)«Ō7+
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{æª~tð~Ōē_İfđ◊Q>~Hāmþn%úKă□đBS;JİümWĐ-2S=·hTşø□Ā™□İ+&÷ççþ%Es¹ðµ□.Çþ·Ō Ůµæ¹[Đēªİwİªzs?
žĒē◊y□İŮŮĐœşŸªª×ôŸyş#ðw□¼ăœ¹şŞ¼¼ªœŌ(þþİFYĐăžİ_±ŌŮ) İª¹9◊FŮT%◊◊#TGlăăĒŮ:¹ZŌBçrĐşş#Mİ,ÇKğ%#İçô/
ôœ□◊◊İŸ□□Ů>~£M-Āú□◊ªç9Ē ŮŮşµ4gô>£çô~ç²/□¼k7ŸðÇþŸ·ăđŮ□□þW□—đİĐŌx·İ~ĀWp~£WwđİŌĒ·Æİİj□žŌİT
"ē#tþİªİŌİŮİ+ēd+Gj□w~İG&ŸŮ²ôŮµ™v £□şđŮŸ%Ÿ±÷÷žēăŸ—İŮŃİ?dkŮôşçg#Ů□İ"ŮđŮ%◊)İēCİĐ:
ª·7WİB7□□Āİİ=□◊Ů~4İ,ª@µ&"ž+İăřx±ĐŸĐªş%VHÇ±U>vŸŸŸKĀ◊+Āăİ¹□·W,"ô5ôēkž—«æsWđG1)□◊B»ŃZ#İē+F·Vnª³Ōİ»ş\$ât·ĐŮŮŸŮ
—Ā+◊□Şē◊đòİ:İ◊"£;ăjİÆİ&Āă·ŮĀ¾¼µçşæHŌ^İj7EŸİĀQ4#~V×İŮ%6□Ńăª~eM"ôşF"—
ŌŌİ□÷İ^ĀŮŸŌdÆX+□◊□,Ů□ĐVUŮ3j)—İ~úİçŸ½Ēē~«ēŮŌ:ŸİŮİ[V~œ37đòİv?þfĒ&TUðŮ²ð±~şÇžēăk~žİşzSL&□-
<~„İ!~Ůþô÷÷...f¹m'□İþİµ=□1ĐvOZ~◊İPôŌŮB <ēĐg™ryŮ~B-ĒİĀİ&ăŸZÆşô"◊□cNİĀăİŮç¹Ōžž~V¹Ń¾ª±ăµµ_Ş
~9éª5½Z-wŮ½İ;Đ,Āđ◊9ĐUŮ÷CŌW9×ŮŮ²đFİ¼·ĐĀ~ăwĐēŌž...◊ÉU:~x&ŸZ ĐM#jİİ)...³uĐ.MJ
&GzŃēĐUŮŸ½T~?□◊~ªª%◊ē□□İ¹İŮœv/
Ē~ēþH+×%◊ªŮ~Ē~ª¹ĀĀă÷ªª◊ĒēĐUJ'ĐPşmZ†İlX·ĐW...ç\$Ēăk□ŌMŌĒēēē□□žêRĐ'Ēôªēİ.[X-
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€Ō◊:□□_□□◊P+78İnİj◊ªŸ½UUUjİªª¼ŌİŮşç£†7İ#şşēŌŞç9ē²~"UŮ◊"◊^Ÿç·ĐĀĐUŮK#~
cQ7UU^HŸ9Eu%:XDŞ°M_E/9İ¹◊%◊ñnİăŌžĐD_&TŌž—Z³wĐxžĐ□□FŮđµ¼¼◊µŮŮs)ŸªþĐŸç<³9vB2¾çcŌ={½wæ^Ů□-
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ñžŮ"ŮôĒçÇ#~:Ō²æ²,µŮ„İİ·CēİĀwİ[Æ8qđŌŃfŮ5"□◊İ¹e¹¼c†fŮ½²þ1Ā¹ŸŸĐ9~va;Ů-ăđ;ŮŮİ-ñžŮŮtŮĀİē□k³
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ăđ;ŮŮİ-ñžŮŮtŮĀİē□k³ þŮŮŮăŮ„İİ·Cēă3şđ□□k(+)ŮôĒçÇĐİ¹e¹¼c†fŮ²ĐŮşYAZİĀwİ[Æ8va;Ů-ăđ8CEēőĀfşĒ×fŮ½²þ1Ā³
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α.èÙKdgÖNÇKú[9ΠÖZ±ñ5P·Ö[VÖèQEni...Ê:αΠC9~δ-u.ΠÈ\`È2½ΠeòRÖ³©çíM\$íð)Πílbck½[4·²Π:7zñΠ...ΠzÖ1—
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Üª±èqΠÖKRTTý&íDDäBdM,,ZÄjèÿ÷~¹~4;mi¾4œμN`fW`ΠÄö[Π·ly;²Ztv\$±v)\ÇHª~k·κkÄÝsŠžŠ&émfΠ`š1ªÉ1Πvã¥...
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äš,Π~ΠNíæ³IÖñ·ÈXè««T,u[t«ΠitZ·Π·fb""FY2úÖòòΠ]pØO²&yÈ>½Ytí(Šäc×èüü·š©ΠTEè
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1ø£ÇÈÇNK`)^S«»Ü òDBùCÈuw{a>È:ÿ(p·o<<|CEtæ°,·ä:»½°YdMý`<\$Ww1Πi%ò¿ò² 1ø£ÇÈÇNK`)^S«»Ü
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žÌ,·K~>RΠDEfJÆÈÖwΠDStÜΠΠD:ó%ò%òYÖi[ØÖ Π:ó%ò%òYÖi[ØÖΠòGàÚj]šìégì.€Π4€ΠffpΠ`ú»I23AüŽÄj)]í·pYjYáz»Pyò;/
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;klD`&×ΠU2òèHÈm`ΠY`#Uªæç;èòj¢t8u)4p·Ú»ΠÓžΠ×™l8;Πeš·#5`\$`¥Ufj£Πúyø`.,o6úŽÝÜ`\*#Ç©ìÜÖΠΠæ~ò`©4
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æE:ΠΠj`^²÷Π`làòZΠ£?`fYšb= Nl|°Š«×YviΠò-4ªTix`ÈΠÖ±zÆ*ÈΠcvM¹)/
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²ØQj;ΠYÖÜHÜHÜòBèl¹Z`»FciQYDFÚ5³>|Πæjž9àTΠ£Ömí·~p—`ðúcZΠ,ñ™®äÖ%ò,pq
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j)pDp°;çjΠKΠD^ò`MÖ~`ìΠRN?μÜΠ)p³ò3W9Ö·ΠjØΠiSΠ¢²#ªèU`†{ò:~ªªðb(-~VóV=Èª¿«†èl™¿Ö=:SéJnŽòt2~`Rè
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Figure 1 – UML Use Case Diagram of the Facility Reservation & Approval System. Editable sources: [usecase-diagram.puml](#) (PlantUML), [usecase-diagram.mmd](#) (Mermaid).

UC-02 "Submit Reservation Request"

Field	Description
ID	UC-02
Actor	Requester (Student / Faculty / Student Org); Administrator may submit on behalf
Precondition	Requester is logged in; at least one Active facility exists
Trigger	Requester opens the Reservations page and enters facility, purpose, participants, start, end
Main flow	1. System lists Active facilities and their capacity. 2. Actor selects facility, enters purpose, participant count, start, end. 3. System validates start < end (BR-B4-02), capacity ≥ participants, overlaps vs Approved / Scheduled / In Use (BR-B4-03), and facility Active (BR-B4-01). 4. System determines route: Auto-confirm → status Approved; Requires approval → Pending; >100 pax → dual approval (CR-01). 5. System stores the reservation, logs (BR-B4-10), notifies the requester (UC-09).
Alternatives	5a. Auto-confirm: immediate Approved + notification. 5b. Large event: status Pending / Partially Approved and Security / Admin notified (UC-10).
Exceptions	Validation failures (overlap, capacity, inactive facility, inverted time) return a specific blocked message and no record is created.
Postcondition	Reservation saved in the system; requester notified; audit entry written.

Field	Description
ID	UC-03
Actor	Administrator (Security for large-event leg)
Precondition	Request exists with status Pending (or Partially Approved)
Trigger	Administrator opens the Approval Queue
Main flow	1. System lists pending requests (large events for Security). 2. Actor selects Approve or Reject. 3. System re-checks overlap for approval (BR-B4-03). 4. Non-large: status → Approved / Scheduled or Rejected; decision stored. 5. Log audit; notify requester.
Alternatives	3a. Overlap → approval blocked with message. 4a. Large event (UC-10): Admin and Security each decide; both Approved → Approved; either Rejected → Rejected; mixed → Partially Approved.
Exceptions	Terminal reservations (Completed / Rejected / Cancelled / No-show) cannot be decided; rejected cannot go to Scheduled (BR-B4-05).
Postcondition	Status updated, requester notified, audit entry written.

31. Requirements Traceability Matrix

Requirement	Source	Use Case(s)	Business Rule	Test Case	Result
FR-01	S1	UC-01	BR-B4-01	TC-B4-08 (inactive / maintenance blocked)	Passed
FR-02	S1	UC-02	BR-B4-02, BR-B4-03	TC-B4-01, TC-B4-02	Passed
FR-03	S3	UC-03, UC-04	BR-B4-05, BR-B4-06, BR-B4-07	TC-B4-05, TC-B4-06	Passed
FR-04	S2	UC-02, UC-03	FR-04 policy	TC-B4-01 (auto-confirm)	Passed
FR-05	S2	UC-03	BR-B4-04	TC-B4-03, TC-B4-04	Passed
FR-06	S3	UC-04	BR-B4-07, BR-B4-09	TC-B4-05, TC-B4-06, TC-B4-07	Passed
FR-07	S4	UC-06	after-hours rule	Manual (Auth List tab)	Passed
FR-08	S5	UC-05	â€”	Manual (Reports + CSV export)	Passed
FR-09	S1â€”S4	UC-02, UC-03, UC-09, UC-10	BR-B4-10	Manual (Notifications tab)	Passed
FR-10	S2	UC-07, UC-08	RBAC matrix	TC-B4-09 (audit), RBAC checks	Passed
CR-01	S4	UC-10	CR-01 (> 100)	TC-CR-01	Passed
BR-B4-10	S6	all	audit trail	TC-B4-09	Passed

32. Change-Request Impact Analysis

CR-01 â€” Dual Approval for Large Events (>100 participants)

Aspect	Analysis
Rationale	Security raised a concern that events >100 pax need a second, security review before approval.
Affected requirements	FR-03 (workflow), FR-05 (approval), FR-06 (state handling); new UC-10; new status Partially Approved .
Affected components	Approval logic (<code>decideApproval</code>), transition map, approval-queue UI filter for Security, reservation form (participant-count gate); schema gains <code>admin_decision</code> , <code>security_decision</code> , <code>is_large_event</code> .
Affected business rules	Adds CR-01 alongside BR-B4-04 · 05; Partially Approved becomes a legal state.
Test impact	New test TC-CR-01 (150 pax → Partially Approved → Approved after both decision-makers act).
Risk / effort	Medium; low technical risk since it reuses the existing review flow with an added second sign-off; requires Security login and queue visibility.
Decision	Approved â€” merged in current release; threshold configurable via <code>LARGE_EVENT_THRESHOLD = 100</code> .